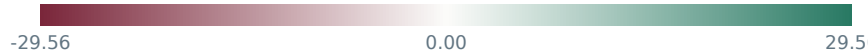
Routed timing: internal paths and complete design

Routed OOC | Vivado 2024.2 | xc7a35tcsg324-1 | 20 ns target

One signed nanosecond scale; zero marks the corresponding check boundary.

	Overall setup WNS	Register setup slack	Hold WHS	Pulse width WPWS
Original core	**-0.884** 1 / 5,944 fail	**+0.058** filtered worst path	**-2.232** 131 / 5,944 fail	**+9.500** 0 / 1,962 fail
With isqrt	**-29.556** 2,970 / 5,954 fail	**-29.556** filtered worst path	**-2.168** 128 / 5,954 fail	**+9.500** 0 / 1,966 fail



Slack (ns): negative = violation; nonnegative = check met

Original / with isqrt: setup fail / fail; hold fail / fail.

Register setup uses a separate path filter; it does not supply endpoint failure counts.

External clock context and interface routing are incomplete. No implemented Fmax is claimed.